Finite positivity forces binomial rigidity

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Task 1: complete classification for every positive integer s

Let $s \geq 1$ be an integer, $q \in \mathbb{R} \setminus \{0\}$, and $t = \widetilde{q} > 0$. For real $p_0, \ldots, p_s$, define

$$g_k = \sum_{m=0}^{\min(k,s)} \binom{s-1+k-m}{s-1} p_m q^m (-t)^{k-m} \qquad (k \geq 0). \tag{1}$$

Lóczi and Ketcheson's Conjecture 2.11 in the preprint, numbered 8.2 in the published paper [1], asks whether $g_0, \ldots, g_{s+1} \geq 0$ force the binomial coefficients below. We prove this for every s, and give the exact interpretation of the extra variable p_{s+1} printed in that conjecture.

Theorem 1. *The following conditions are equivalent:*

1. $g_k \geq 0$ *for* $0 \leq k \leq s+1$*;*
2. *for some* $c \geq 0$*,*

$$p_m = c\binom{s}{m}\left(\frac{t}{q}\right)^m \qquad (0 \leq m \leq s).$$

In either case $c = p_0$*,* $g_0 = c$*, and* $g_k = 0$ *for every* $k \geq 1$*. If* p_{s+1} *is also supplied as an independent real variable, it is arbitrary: the complete feasible set is the displayed ray in* $(p_0, \ldots, p_s)$ *times* $\mathbb{R}$ *in the last coordinate.*

Proof. All identities can be read in the formal power series ring $\mathbb{R}[[z]]$; no convergence assertion is needed. Set

$$P(z) = \sum_{m=0}^{s} p_m q^m z^m, \qquad G(z) = \frac{P(z)}{(1+tz)^s}.$$

The negative binomial expansion

$$(1+tz)^{-s} = \sum_{j \geq 0} (-1)^j \binom{s+j-1}{s-1} t^j z^j$$

shows that $G(z) = \sum_{k \geq 0} g_k z^k$, with precisely the sums in (1). Since $\deg P \leq s$, the coefficient of z^{s+1} in $(1+tz)^s G(z) = P(z)$ is zero. Thus

$$0 = \sum_{j=0}^{s} \binom{s}{j} t^j g_{s+1-j}. \tag{2}$$

Every weight $\binom{s}{j}t^j$ is strictly positive. Under condition (1), every summand is nonnegative, so

$$g_1 = g_2 = \cdots = g_{s+1} = 0.$$

Also $g_0 = p_0 \geq 0$. Comparing coefficients of degrees 0 through s in $(1+tz)^s G = P$ now gives

$$p_m q^m = g_0 \binom{s}{m} t^m \qquad (0 \leq m \leq s).$$

Division by q^m proves (2), including negative q. Conversely, (2) gives $P(z) = c(1+tz)^s$, so $G(z) = c$ and all the asserted values of g_k follow. Finally, p_{s+1} occurs in none of the defining sums, so it is unrestricted. □

The printed last index. Both source versions also assert the formula for $m = s+1$, which would force $p_{s+1} = 0$. That extra conclusion is false when p_{s+1} is independent: take $s = q = t = 1$ and $(p_0, p_1, p_2) = (1, 1, 1)$. Then $(g_0, g_1, g_2) = (1, 0, 0)$. Theorem 1 proves the full intended rigidity assertion for all the coefficients that enter the hypotheses; it also proves every claimed vanishing sum and its converse. The complete all-s classification, rather than this unused-coordinate counterexample alone, is the mathematical result here. In the motivating degree-s polynomial setting, $p_{s+1} = P^{(s+1)}(x)/(s+1)! = 0$ automatically.

Proposition 1 (A general denominator and a sharp cutoff)**.** *Let $D(z) = \sum_{j=0}^{s} d_j z^j$ with every $d_j > 0$, and $P \in \mathbb{R}[z]$ with $\deg P \leq s$. If the coefficients $g_0, \ldots, g_{s+1}$ of P/D are nonnegative, then $P = g_0 D$. For each such D, testing only degrees $0, \ldots, s$ does not imply this conclusion.*

Proof. The coefficient of z^{s+1} gives $\sum_{j=0}^{s} d_j g_{s+1-j} = 0$. Positivity again forces $g_1 = \cdots = g_{s+1} = 0$, and comparison through degree s gives $P = g_0 D$. For sharpness, fix $\varepsilon > 0$ and put

$$P(z) = D(z)(1+\varepsilon z) - \varepsilon d_s z^{s+1}.$$

This has degree at most s, and

$$\frac{P(z)}{D(z)} = 1 + \varepsilon z - \frac{\varepsilon d_s z^{s+1}}{D(z)}.$$

Hence $g_0 = 1$, $g_1 = \varepsilon$, and $g_2 = \cdots = g_s = 0$, where the last range is empty if $s = 1$. But $g_{s+1} = -\varepsilon d_s / d_0 < 0$, and P/D is not constant. □

Verification and scope. The exact verifier checks the positive linear certificate by coefficient cancellation, compares direct sums with formal series division, and checks the feasible family and the sharp-cutoff construction. The proof for arbitrary s is the coefficient argument above, not an extrapolation of these computations. The source's Runge–Kutta optimal-radius questions are separate: the sign $t > 0$ used here must not be replaced by $t = -a < 0$ without a new argument.

Four proposed follow-up tasks

Task 2. For denominators having zero or negative coefficients, characterize when some finite number of nonnegative coefficients of P/D, with $\deg P \leq \deg D$, forces P/D to be constant.

Task 3. Determine optimal quantitative bounds on the distance from P to a multiple of D when the tested coefficients are only bounded below by small errors; study uniformity as the positive coefficients of D approach zero.

Task 4. For $\deg P > \deg D$ and a specified finite testing range, classify the extreme rays of the cone defined by coefficient nonnegativity of P/D.

Task 5. Find a replacement for the positive certificate (2) compatible with the denominator $(1 - az)^s$ and the order conditions in the source's general rational-function optimal-radius conjectures.

These are proposed research directions and receive no additional solution credit.

References

[1] L. Lóczi and D. I. Ketcheson, *Rational functions with maximal radius of absolute monotonicity*, LMS Journal of Computation and Mathematics **17**(1) (2014), 159–205. Conjecture 8.2, p. 197. doi:10.1112/S1461157013000326. Preprint: arXiv:1303.6651v1, Conjecture 2.11, p. 10.